ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PRACTICE EXAM 1 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 7 Natural Sciences practice exam 1.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples.

SECTION A

QUESTION 1

1 1.1 D ✓ 1.2 B ✓ 1.3 D ✓ 1.4 B ✓ 1.5 C ✓ 1.6 A ✓ 1.7 B ✓ 1.8 D ✓ 1.9 A ✓ 1.10 C ✓ (10)

QUESTION 2

2 2.1 Biosphere ✓ 2.2 Pollination ✓ 2.3 Invertebrate ✓ 2.4 Variation ✓ 2.5 Indicator ✓ 2.6 Element ✓ 2.7 (Heat) insulator ✓ 2.8 Kinetic energy ✓ 2.9 Phases of the Moon ✓ 2.10 Orbit ✓ (10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1.1 A petal ✓ B anther ✓ D stigma ✓ F ovary ✓ (4)

3.1.2 B ✓ and C ✓ (anther and filament) (2)

3.1.3 Any ONE: contains the ovules (female sex cells) ✓; develops into the fruit after fertilisation (1)

3.2 Any TWO: large, brightly coloured petals ✓; a sweet scent ✓; nectar; sticky or spiky pollen; landing platform for insects (2)

3.3 Pollination is the transfer of pollen from an anther to a stigma ✓. Fertilisation is when the male sex cell (from the pollen) joins with the female sex cell (egg) in the ovule ✓. (2)

3.4 (a) Animals ✓ (b) Wind ✓ (c) Water ✓ (3)

3.5 (a) Amphibian ✓ (b) Mammal ✓ (c) Reptile ✓ (3)

3.6 (a) Any TWO: growth spurt ✓; hair grows under the arms and in the pubic area ✓; skin becomes oily / pimples; sweat glands become more active; mood changes (3)
(b) Fallopian tube (oviduct) ✓

[20]

QUESTION 4

4.1 Use a magnet to remove the iron filings ✓. Add water to the salt and sand and stir so that the salt dissolves ✓. Filter the mixture ✓; the sand stays behind on the filter paper ✓. Evaporate (heat) the salt water (filtrate) so that the water evaporates and the salt is left behind ✓. (5)

4.2 (a) Chromatography ✓ (b) Black ink is a mixture of different coloured inks (pigments) ✓ (2)

4.3.1 A acid ✓ B base ✓ C neutral ✓ (3)

4.3.2 Any ONE: wear safety goggles / gloves ✓; never taste or smell chemicals directly; add acid to water, never water to acid; rinse spills with lots of water (1)

4.4.1 O ✓ Fe ✓ Ca ✓ (3)

4.4.2 Potassium ✓ Copper ✓ (2)

4.4.3 The left side (and the middle) ✓ (1)

4.4.4 Any TWO: good conductors of heat ✓; good conductors of electricity ✓; shiny (lustrous); malleable and ductile; high melting points; strong (2)

4.5 Metal conducts heat to cook the food, but plastic or wood is an insulator so the handle stays cool enough to hold ✓ (1)

[20]

QUESTION 5

5.1 (a) Radiation ✓ (b) Conduction ✓ (c) Convection ✓ (3)

5.2 The water at the bottom is heated, expands and becomes less dense, so it rises ✓. The cooler, denser water at the top sinks to the bottom to be heated ✓. This convection current carries heat through all the water ✓. (If the element were at the top, only the top water would heat.) (3)

5.3 (a) Any TWO: install a ceiling or roof insulation ✓; seal gaps around doors and windows ✓; use thick curtains; carpets or rugs on the floor; double-glazed windows (3)
(b) The ceiling (and insulation) traps a layer of air, which is a poor conductor, so less heat escapes through the roof ✓

5.4 (a) Chemical potential energy ✓ → electrical energy ✓ → light energy ✓ (+ heat) (5)
(b) Light is useful ✓; heat is wasted ✓

5.5 Coal is burned to release heat ✓, which boils water to make steam ✓. The steam turns a turbine ✓, which turns a generator that produces electricity ✓. (The electricity is sent out through transformers and power lines.) (4)

5.6 Any TWO: coal is non-renewable and will run out ✓; burning coal releases carbon dioxide (global warming) ✓; air pollution and acid rain; mining damages the land; power stations use a lot of water (2)

[20]

QUESTION 6

6.1.1 Winter ✓ (1)

6.1.2 At A the Southern Hemisphere is tilted away from the Sun ✓, so the Sun's rays reach it at a low angle and are spread over a larger area, and the days are shorter ✓. (2)

6.2 (a) Day and night ✓ (b) A year (and, with the tilt, the seasons) ✓ (2)

6.3 (a) It reflects light from the Sun ✓ (6)
(b) New moon ✓ → first quarter ✓ → full moon ✓ → last (third) quarter ✓
(c) About 29½ days (about one month) ✓

6.4 The gravitational pull of the Moon ✓ (and the Sun) on the water in the oceans ✓ (2)

6.5 The Moon moves between the Sun and the Earth ✓ and its shadow falls on part of the Earth, blocking the Sun's light ✓. (2)

6.6 Ptolemy: the Earth is at the centre and the Sun and planets move around it (geocentric) ✓. Copernicus: the Sun is at the centre and the Earth and planets move around it (heliocentric) ✓. (2)

6.7 He saw moons orbiting Jupiter, so not everything moves around the Earth ✓; he saw that Venus has phases, which can only happen if Venus moves around the Sun ✓. (2)

6.8 Any ONE: SALT (Southern African Large Telescope) at Sutherland ✓; MeerKAT / SKA radio telescope in the Karoo (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE